

VIVEKANANDA INSTITUTE OF PROFESSIONAL STUDIES - TECHNICAL CAMPUS

Grade A++ Accredited Institution by NAAC

NBA Accredited for MCA Programme; Recognized under Section 2(f) by UGC;
Affiliated to GGSIP University, Delhi; Recognized by Bar Council of India and AICTE
An ISO 9001:2015 Certified Institution

SCHOOL OF ENGINEERING & TECHNOLOGY

ASSIGNMENT 1 (UNIT-I & UNIT-II)				
SUBJECT: Distributed Systems and Cloud Computing		SUBJECT CODE:CIE-407T		SEM: VII BRANCH: CSE
S. NO	QUESTION	CO	BL	MARKS
1	Define a Distributed System and discuss its key characteristics. Explain different examples of distributed systems such as Client-Server, Peer-to-Peer, Grid, and Cloud Computing systems.	1	1,2,3,4	5
2	What are System Models in Distributed Systems? Explain the Architectural Models (such as layered architecture, object-based architecture, and event-based architecture) and Fundamental Models (interaction, failure, and security models).	1	1,2,3,4	5
3	Explain the concept of Time and Global States in distributed systems. Discuss in detail the working of Lamport's Logical Clocks and Vector Clocks?	2	1,2,3	5
4	What is the Consensus Problem in Distributed Systems? Explain the challenges in achieving consensus and describe algorithms or mechanisms used for Distributed Mutual Exclusion.	2	1,2,3,4	5